

Project Design (Updated for CSE499B Final Submission):

AI Medical Pre-Screening Assistant

From a 5-Stage ASR Pipeline to a 15-Module Conversational Clinical Intelligence System

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1 Introduction

This chapter presents the updated project design of the *AI Medical Pre-Screening Assistant*, a system designed to enable patients — particularly in Bangladesh — to describe their symptoms verbally in Bangla, Banglish, or regional dialects, and to receive a structured clinical pre-screening report that the attending physician can review before the consultation begins.

The design presented here is a significant evolution from the five-stage pipeline described in the original CSE499A proposal. Based on empirical benchmarks completed during Phases 1–3 of the project, feedback from the faculty advisor, and a critical review of clinical requirements, the system has been redesigned from a passive transcription pipeline into an active, conversational, multi-module clinical intelligence system. The remainder of this chapter explains what changed, why it changed, and how the new architecture is organised.

The design is presented in three layers: (i) the updated 15-module system architecture that turns a patient’s spoken complaint into a structured, physician-ready pre-screening report; (ii) the hardware and software components used to realise the pipeline; and (iii) the implementation methodology across the five operational phases of CSE499A.

Important disclaimer: This system is a pre-screening and decision-support tool, not a diagnostic system. All outputs are intended to assist qualified physicians, not replace clinical judgement. Emergency detection flags are alerts, not diagnoses. Risk levels are probabilistic classifications, not medical determinations. Final medical decisions remain entirely with the attending physician.

2 Evolution from the Previous Design

2.1 What the original design was

The original CSE499A design described a five-stage linear pipeline:

1. **Patient Speech Capture** — voice recorded on a tablet or kiosk.
2. **Automatic Speech Recognition (ASR)** — Bangla-tuned transcription of the recording.
3. **Medical Named Entity Recognition (NER)** — extraction of symptoms, medications, durations, allergies, and anatomical sites from the transcript.
4. **Structured EHR Construction** — packaging of labelled spans into a structured record.
5. **Doctor Dashboard with Differential Hints** — physician review interface displaying the record and lightweight diagnostic hints.

This was a single-pass, stateless pipeline: the patient spoke once, the system transcribed and extracted once, and a report was generated from that single input. There was no mechanism for detecting missing information, no risk stratification, no follow-up dialogue, and no emergency triage pathway.

2.2 What the empirical results revealed

The Phase 1–3 benchmarks demonstrated that even the strongest open-source Bangla ASR model (BengaliAI Regional) achieves only 46.94% WER on real dialect speech, meaning that approximately half of all words are mis-transcribed. Larger multimodal audio language models (1.7B–7B parameters) performed significantly worse. This finding has a direct consequence for system design: if the upstream transcript is noisy, a single-pass extraction pipeline will silently miss clinical information with no mechanism to recover it. A patient who mentions a critical symptom in a poorly-recognised segment will have that symptom absent from the report.

This realisation, combined with clinical requirements gathered during Phase 5 (comparative chatbot study), led to the conclusion that the system must be *interactive*, not passive. Rather than extracting once from a noisy transcript, the system should identify what is missing and ask targeted follow-up questions to fill those gaps.

2.3 Key differences between old and new design

Table 1 summarises the principal structural differences between the original five-stage design and the updated fifteen-module design.

Table 1: Comparison of original and updated system designs.

Dimension	Original 5-Stage Design	Updated 15-Module Design
Interaction model	Single-pass, stateless. Patient speaks once; system processes once.	Conversational, stateful. System identifies gaps and conducts targeted follow-up dialogue.
Missing information handling	Not addressed. Silent omissions passed to the doctor.	Explicit gap detection (Module 6) followed by dynamic follow-up generation (Module 7) in a feedback loop (Modules 8–9).
Emergency detection	Not present.	Dedicated parallel module (Module 5) with rule-based and AI classifier; immediate escalation pathway.

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Table 1 continued

Dimension	Original 5-Stage Design	Updated 15-Module Design
Risk stratification	Not present (only differential hints).	Four-tier risk assessment (Low / Medium / High / Critical) with weighted clinical rules (Module 10).
Explainability	Not present.	Dedicated XAI module (Module 11) generating plain-language explanations of risk assignment (SHAP/LIME or rule traces).
Clinical report	Structured EHR record with differential hints.	Full pre-screening report: chief complaint, symptoms, emergency flags, F/U Q&A, risk level, possible condition categories, recommended next steps.
Language coverage	Bangla and Bangla–English code-mixed.	Bangla, Banglish, regional dialects. Follow-up questions generated in Bangla or English as appropriate.
Feedback loop	Not present.	Module 15 collects structured doctor feedback for continuous model retraining.
Auditability	Audio traceable to transcript; transcript visible on dashboard.	Full session stored in EHR (Module 13): transcripts, extracted data, follow-up exchanges, risk assessments, reports; versioned overrides.

3 Updated System Architecture: 15-Module Pipeline

3.1 Architectural overview

The updated system is organised as a fifteen-module pipeline. The pipeline is no longer strictly linear: Module 5 (Emergency Detection) runs in parallel with the main flow; Modules 7–9 form a conditional feedback loop; and Module 13 (EHR Database) is a persistent store written to at multiple points. Figure 1 shows the complete flow.

3.2 Module 1: Speech-to-Text

Goal: Convert patient voice input into raw text.

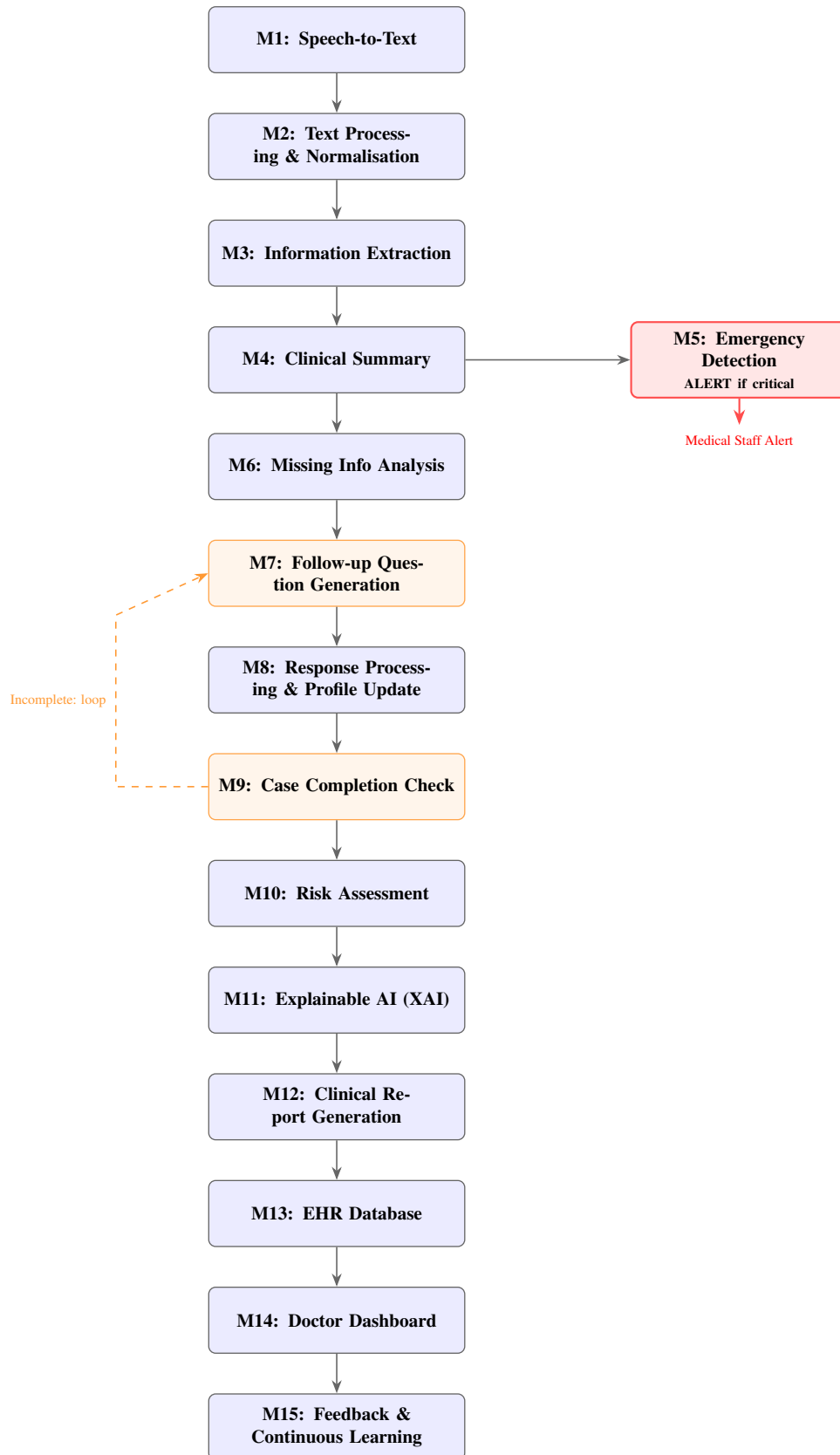


Figure 1: Complete 15-module pipeline of the AI Medical Pre-Screening Assistant. Orange loop indicates the follow-up question cycle; red branch is the emergency escalation pathway.

The patient speaks freely in Bangla, Banglish, or a regional dialect. A speech recognition model trained or fine-tuned on Bangladeshi speech patterns handles transcription. The module must tolerate ambient noise, strong accents, and code-switching between Bangla and English. If speech quality falls below a configurable threshold, the system falls back to a manual text-entry interface.

Key requirements: Dialect-aware acoustic model; real-time or near-real-time transcription; noise-tolerant voice activity detection; fallback to manual text input.

Connection to Phase 1–4 benchmarks: The empirical work completed during CSE499A directly feeds this module. BengaliAI Regional (46.94% WER) is the current strongest baseline; fine-tuning Qwen3-ASR-1.7B with LoRA on the multi-dialect corpus (scheduled for CSE499B) is the pathway to a production-grade transcription engine.

3.3 Module 2: Text Processing and Normalisation

Goal: Clean, standardise, and prepare raw transcribed text for downstream analysis.

This module handles spelling correction, normalisation of Banglish text, removal of filler words, punctuation restoration, and sentence boundary detection. Where necessary it translates Bangla or Banglish content into a standardised form that the rest of the pipeline can process consistently.

Key requirements: NLP preprocessing pipeline; Bangla–English bilingual handling; medical spelling correction; noise and artefact removal.

3.4 Module 3: Information Extraction

Goal: Extract structured clinical data from the normalised text.

The system identifies and tags key clinical entities including symptoms, affected body parts, duration, severity, frequency, patient age, and any pre-existing conditions or medications mentioned. This forms the foundation of the patient profile.

Key requirements: Medical NLP; named entity recognition for clinical entities; temporal expression extraction (e.g., “for three days”, “since last week”).

Connection to original design: This module directly extends the Medical NER stage from the original five-stage pipeline, expanding the entity set and adding temporal reasoning.

3.5 Module 4: Initial Clinical Summary Generation

Goal: Generate a concise, human-readable summary of the patient’s chief complaint from the extracted entities.

Example output: *“Patient reports fever and dry cough for 3 days, with mild fatigue. No mention of breathing difficulty or chest pain.”*

Key requirements: Medical text summarisation model; templated and generative output options.

3.6 Module 5: Emergency Detection

Goal: Immediately flag life-threatening or urgent conditions before the full pipeline runs.

This module runs in parallel with or immediately after extraction. It checks for red-flag symptoms — chest pain, stroke indicators, severe breathing difficulty, loss of consciousness, acute trauma — and triggers an immediate alert if any are detected. This module has absolute priority over all others.

Key requirements: Rule-based critical symptom checker; AI classifier for ambiguous emergency presentations; escalation pathway to alert medical staff immediately.

New in updated design: Emergency triage was completely absent from the original architecture. Its addition was driven by clinical safety requirements identified during the Phase 5 chatbot study.

3.7 Module 6: Missing Information Analysis

Goal: Identify clinical information gaps that are necessary for a complete pre-screening profile.

Using clinical reasoning rules and medical knowledge, the system checks what information is present versus what is expected and produces a structured gap report.

Example output:

- ✓ Symptom detected: Fever
- × Temperature reading not provided
- × Duration of fever not specified
- × Breathing difficulty not assessed

Key requirements: Symptom completeness logic; clinical checklists per condition category; structured gap report generation.

New in updated design: This module is the principal motivation for moving from a single-pass to a conversational architecture. ASR errors in Module 1 mean that a single pass will routinely miss clinical information; Module 6 makes those omissions explicit rather than letting them propagate silently to the physician.

3.8 Module 7: Dynamic Follow-up Question Generation

Goal: Generate targeted, conversational questions to fill the identified information gaps.

Questions are generated in natural language — in Bangla or English as appropriate — prioritised by clinical importance, and adapted to avoid redundancy with information already collected.

Example output: “*Apnar jwor ki matro kal theke? Thermometer diye measure korechen?*”
/ “*When did your fever start? Have you measured your temperature?*”

Key requirements: Question generation model with medical context; Bangla/English bilingual output; priority ranking of questions.

3.9 Module 8: Response Processing and Profile Update

Goal: Analyse patient answers to follow-up questions and update the patient profile.

New answers go through the same extraction and normalisation pipeline as the original input (Modules 2–3). Extracted information is merged into the existing patient profile, resolving contradictions and marking newly filled gaps.

Key requirements: Incremental NLP pipeline; patient profile state management; conflict resolution logic.

3.10 Module 9: Case Completion Check

Goal: Determine whether sufficient information has been collected to proceed to assessment.

The system scores the completeness of the patient profile against a minimum threshold for reliable pre-screening. If critical gaps remain, it loops back to Module 7. The loop continues until the profile is sufficiently complete or a configurable maximum number of turns is reached (to avoid patient fatigue).

Key requirements: Completeness scoring model; configurable thresholds per condition type; loop management and turn limits.

3.11 Module 10: Risk Assessment

Goal: Classify the patient into a risk tier based on their complete profile.

The system assigns one of four risk levels using a combination of clinical decision rules and a trained risk prediction model. Risk factors such as age, comorbidities, symptom severity, and duration are all weighted. Critical-risk cases are re-flagged for immediate attention.

Risk tiers:

Low — Likely minor, self-limiting condition.

Medium — Warrants examination but not urgent.

High — Needs prompt medical attention.

Critical — Requires immediate intervention.

Key requirements: Risk classification model; clinical rule engine; age/comorbidity weighting; integration with Module 5 output.

New in updated design: The original design offered only differential hints; structured, four-tier risk stratification with weighted clinical factors is entirely new.

3.12 Module 11: Explainable AI (XAI)

Goal: Make the system's reasoning transparent and auditable for physicians and regulators.

For every risk assessment, the system generates a plain-language explanation of which symptoms, durations, and risk factors contributed to the classification.

Example output: *"High risk was assigned due to: fever lasting more than 5 days, reported difficulty breathing, and patient age over 65."*

Key requirements: Feature attribution methods (LIME, SHAP, or rule-based reasoning traces); human-readable explanation generation; display on the doctor dashboard alongside the report.

New in updated design: Explainability was absent from the original architecture. Its addition is motivated by medical accountability requirements and by the clinical need for physicians to be able to trust and, where necessary, override automated assessments.

3.13 Module 12: Structured Clinical Report Generation

Goal: Compile all collected information into a formatted pre-screening report ready for the physician.

The report presents a standardised summary that the physician can review before or during the consultation. It does not contain a diagnosis.

Report sections:

1. Patient Profile (age, gender, known conditions)
2. Chief Complaint
3. Symptoms with Duration and Severity
4. Emergency Flags (if any)

5. Follow-up Questions and Answers
6. Risk Level and Explanation
7. Possible Condition Categories (not a diagnosis)
8. Recommended Next Steps

Key requirements: Medical report generation; structured and human-readable output; PDF and digital dashboard export options.

3.14 Module 13: EHR Database

Goal: Securely store all patient data, reports, and interaction history.

Every session — transcripts, extracted data, follow-up exchanges, risk assessments, and final reports — is stored in a structured medical database. Records are linked to patient identifiers and consultation timestamps and support future retrieval, audits, and longitudinal tracking.

Key requirements: Secure, HIPAA/PDPA-compliant database design; encryption at rest and in transit; patient identity management; efficient retrieval by patient ID or date; audit logging.

3.15 Module 14: Doctor Dashboard

Goal: Give physicians a clear, fast interface to review the pre-screening report before or during consultation.

The dashboard displays the structured clinical report, risk level, emergency flags, and the XAI explanation. Physicians can review the conversation history, validate or override extracted information, and add annotations. A notification system alerts physicians to high or critical risk cases immediately.

Key requirements: Web-based dashboard; role-based access control; real-time notifications for high/critical cases; annotation and override tools; mobile-responsive design.

Connection to original design: The doctor dashboard was present in the original design. The updated version adds risk-tier display, XAI explanations, real-time critical alerts, and structured override tools.

3.16 Module 15: Feedback and Continuous Learning

Goal: Use physician feedback to improve system accuracy and clinical relevance over time.

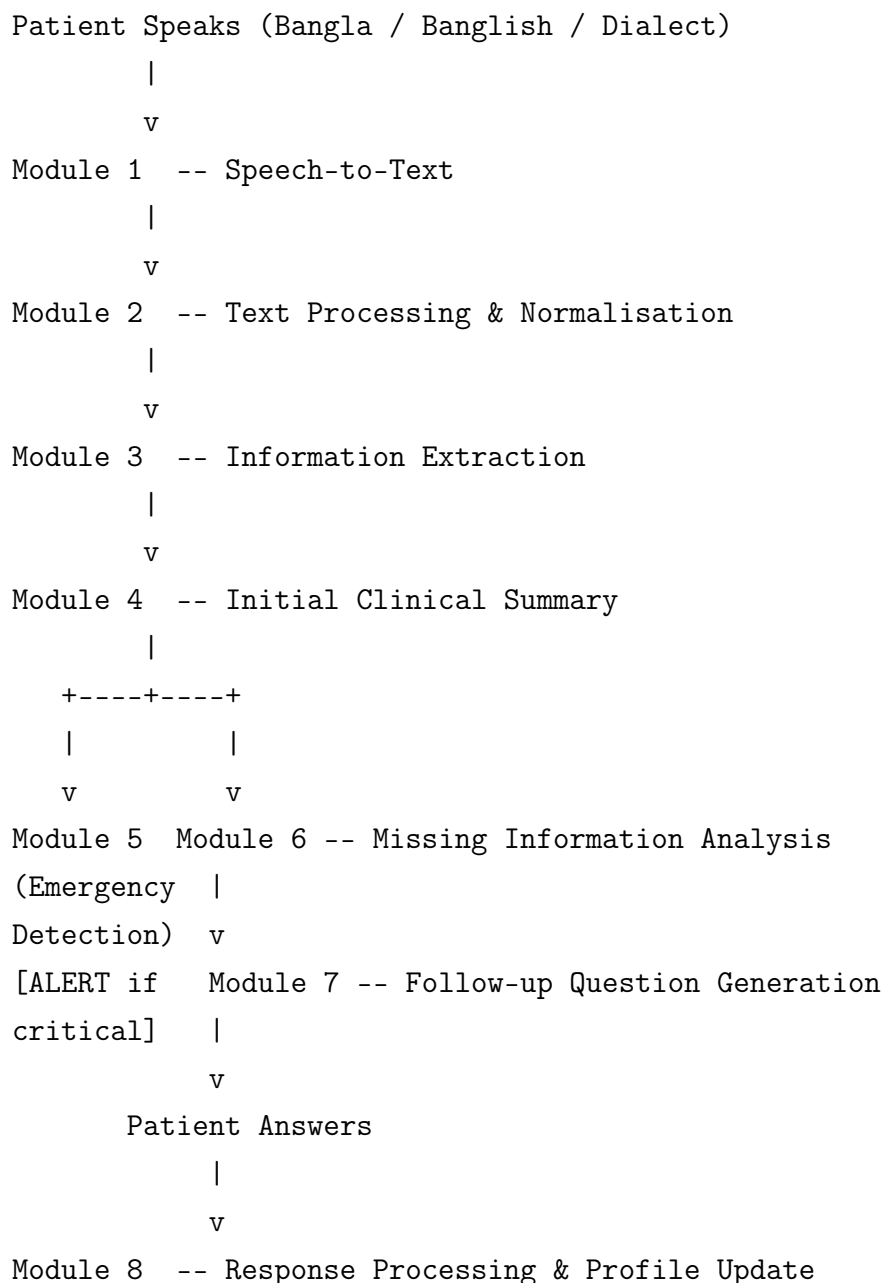
After each consultation, physicians can rate the quality of the pre-screening report, correct inaccurate extractions, flag missed symptoms, and annotate risk-level disagreements. This

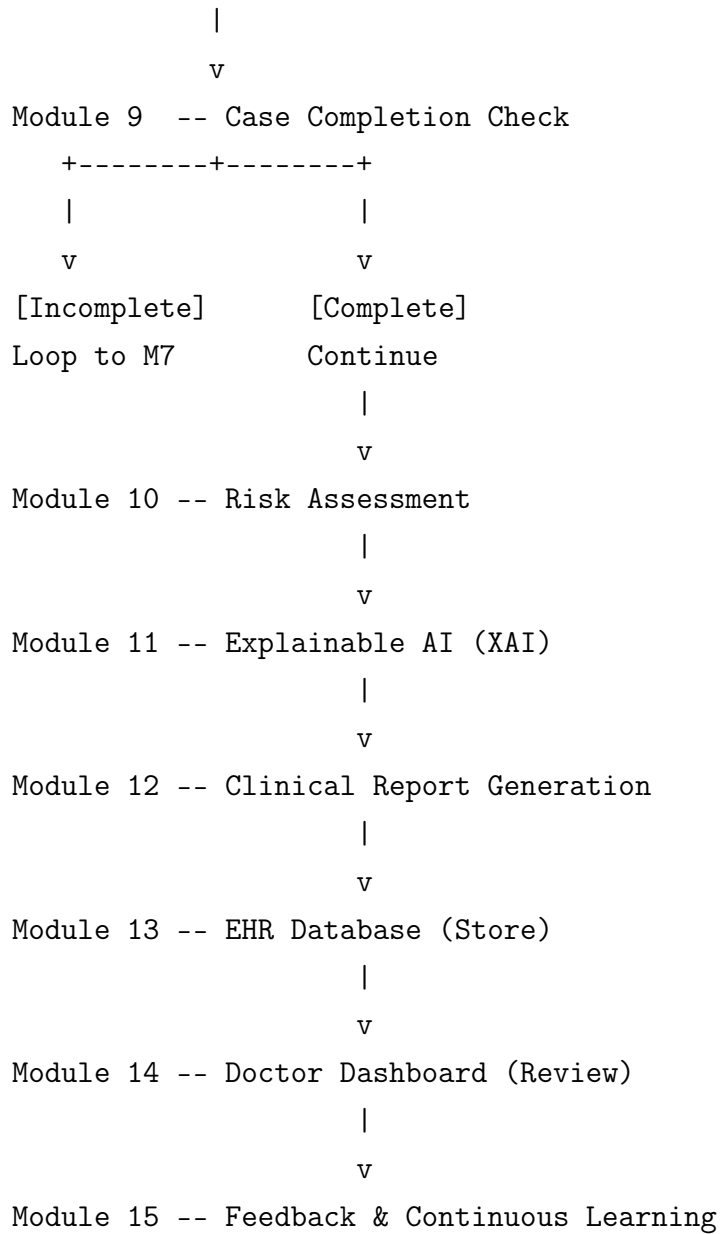
feedback is collected, reviewed, and used to retrain or fine-tune the extraction, risk assessment, and question-generation models.

Key requirements: Structured feedback collection interface; feedback dataset management; model retraining pipeline; performance monitoring and regression testing before deploying updated models.

New in updated design: No feedback loop existed in the original design. Module 15 closes the quality circle and is the mechanism through which the system improves beyond the CSE499B deployment.

4 System Flow Summary





5 Hardware and Software Components

This remains a software-only project; no custom hardware has been fabricated. The patient-facing capture device in deployment would be a consumer Android tablet (8–10 inch form factor) or a fixed clinic kiosk. All model evaluation and development was performed on free-tier Google Colab GPU instances. Table 2 lists the principal software components.

Table 2: Software tools and components.

Tool	Function	Similar tools	Why this one
Python 3.10	Implementation language	R, Julia	Standard for ML/NLP; widest ecosystem.
PyTorch 2.x	Deep-learning framework	TensorFlow, JAX	Native support across all evaluated ASR models; preferred by Hugging Face.
HF Transformers [5]	Model loading, inference, tokenisation	ESPnet, NeMo	Unified API across Whisper, Wav2Vec2, MMS, SeamlessM4T, Qwen-Audio.
HF PEFT / LoRA [6]	Parameter-efficient fine-tuning	Full fine-tune, DeepSpeed	Enables fine-tuning Qwen3-ASR-1.7B on free-tier Colab within memory limits.
BanglaBERT [8]	Medical NER token classification	mBERT, XLM-R	Pre-trained on Bangla; lowest adaptation cost for clinical NER.
spaCy + custom rules	Temporal expression extraction, rule-based clinical checkers	NLTK, Stanford NLP	Fast, production-ready; custom rule layers for Module 6 completeness logic.
Librosa / SoundFile / FFmpeg	Audio I/O, resampling, mono conversion	torchaudio, sox	Stable; runs reliably on Colab.
WhisperX [7]	Long-form audio segmentation	pyannote.audio	Required for audio longer than Whisper’s 30-second window.
jiwer, evaluate	WER, CER, BLEU scoring	sclite	Pure-Python; standard for speech evaluation.
FastAPI (Python)	REST/WebSocket backend	Flask, Django	Async-native; low latency for conversational follow-up loop.
PostgreSQL (encrypted)	EHR store (Module 13)	MySQL, MongoDB	Row-level encryption; relational schema suits structured clinical records.
React / Next.js	Doctor dashboard (Module 14)	Vue.js, Angular	Component ecosystem; server-side rendering for low-bandwidth clinics.
Flutter / PWA	Patient kiosk front-end (Module 1)	React Native	Single codebase; large Bangla font support; single-button record UX.
SHAP / LIME	XAI feature attribution (Module 11)	Integrated Gradients	Model-agnostic; produces human-readable explanations for clinical risk.

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Table 2 continued

Tool	Function	Similar tools	Why this one
Google Colab (T4, A100)	Primary GPU compute	Kaggle, local GPU	Free tier sufficient for all CSE499A inference benchmarks.
Git / GitHub	Source control	GitLab	Required for reproducibility and submission.

5.1 Planned components for the deployed system (CSE499B)

The CSE499B implementation will add the following components above the model pipeline:

- **JWT authentication and role-based access control** — distinct roles for patients (no read access), nurses (intake only), physicians (full record), administrators (no clinical content).
- **PDPA-compliant consent management** — recorded verbal consent integrated into the patient kiosk flow before any recording begins.
- **Model retraining pipeline (Module 15)** — MLflow-tracked experiment management for continuous learning cycles using physician feedback datasets.
- **PDF report export** — formatted clinical pre-screening report (Module 12) exportable as a PDF for printing or EHR attachment.

6 Implementation Methodology

The CSE499A work was organised around five operational phases, each closed by a demonstration to the faculty advisor.

Phase 1 — Audio collection and preprocessing

Demo 1: 12 March 2026. A multi-dialect Bangla audio corpus was assembled from publicly available recordings covering five regional varieties: Pura Dhaka (Dhakaiya), Barishali, Sylheti, Normal Bangla, and Indian Bangla. Each file was renamed under a fixed convention [dialect]_[index]_[gender]_[age].mp3 and logged in dataset_log.csv with source, dialect code, speaker gender, approximate age, and duration. The corpus was preprocessed: resampled to 16 kHz mono PCM, RMS-normalised, and segmented into 10–15-second clips by WhisperX’s voice-activity detector. The final corpus is approximately 4.7 hours, 10 source files, ~1,100 evaluation segments.

Phase 2 — Baseline ASR benchmark (12 models)

Demo 2: 2 April 2026. Twelve open-source Bangla and multilingual ASR models were benchmarked on the corpus: Whisper Small/Medium/Large/Turbo, Wav2Vec2 (XLSR-53, 300M, Common-Voice-Bengali variants), MMS, Vakyansh-Bn, BengaliAI Whisper, BengaliAI Regional, and SeamlessM4T. Each model transcribed the entire corpus and was scored using WER, CER, and corpus-level BLEU. The strongest baseline was BengaliAI Regional at 46.94% WER, 24.29% CER, and 0.273 BLEU. The principal finding was that even the strongest open Bangla model mis-transcribes nearly half of all words on dialect speech — the direct motivator for the conversational gap-filling architecture added in Modules 6–9.

Phase 3 — Larger multimodal audio LLM benchmark

Demo 3: 9 April 2026. Six larger 1.7B–7B-parameter multimodal audio language models were evaluated on the same corpus: Qwen2-Audio (English-routing and direct Bangla modes), Qwen3-ASR-1.7B, Voxtral-Mini, Phi-4 Multimodal, and Qwen2.5-Omni. None matched the smaller specialised baseline; the closest trailed BengaliAI Regional by more than 50 percentage points of WER. The conclusion — “bigger is not better for Bangla dialect speech” — confirms that Module 1 must use a specialised, fine-tuned model rather than a general-purpose large model.

Phase 4 — Qwen3-ASR fine-tuning research (in progress)

Demo 4: 16 April 2026. The team researched the methodology for parameter-efficient fine-tuning of Qwen3-ASR-1.7B on the multi-dialect corpus, using LoRA adapters on the speech encoder with the language-model decoder weights frozen. No training has been performed yet; this phase is a methodology study. The fine-tuning is scheduled for CSE499B and will produce the production ASR engine for Module 1.

Phase 5 — Comparative chatbot output study (selected)

Selected: 23 April 2026; execution scheduled for the next demo. Several public AI chatbots — ChatGPT, DeepSeek, Qwen, Perplexity — will be prompted with simulated patient utterances. Their structured EHR-style outputs will be compared along axes of schema consistency, completeness, handling of code-mixed terms, and source faithfulness. This phase directly informed the design of Module 12 (clinical report schema) and Module 6 (information gap taxonomy), and it surfaced the requirement for emergency triage (Module 5) and explicit risk stratification (Module 10) that were absent from the original design.

7 Data Privacy and Clinical Safety Constraints

Data privacy and consent must be handled before any recording begins. Patients are informed in Bangla of how their voice and health data will be used and stored, and explicit verbal consent is recorded as part of the session (Module 1). All data is processed under HIPAA/PDPA-equivalent constraints: encrypted at rest and in transit, accessible only through role-based controls, and retained for the minimum period required.

Multilingual and dialect coverage will be continuously expanded based on real usage data collected through the feedback loop in Module 15.

Clinical safety boundary: The system outputs a pre-screening report, not a diagnosis. All clinical decisions remain with the licensed physician. The risk levels produced by Module 10 are probabilistic indicators derived from self-reported symptoms; they are not medical determinations and are not legally equivalent to a clinical assessment.

8 Conclusion

The updated project design presented above represents a fundamental redesign of the system from a passive, single-pass transcription pipeline into an active, conversational, fifteen-module clinical intelligence system. The driving force behind this evolution was empirical: the Phase 1–3 benchmarks demonstrated that Bangla dialect ASR at 46.94% WER is too noisy for a single-pass extraction to be clinically reliable, and the Phase 5 chatbot study revealed that multiple critical clinical functions — emergency triage, risk stratification, gap detection, and explainability — were absent from the original design.

The updated architecture addresses these gaps directly. Module 5 provides a dedicated emergency escalation pathway. Modules 6–9 form a conversational loop that closes information gaps before assessment. Module 10 replaces informal differential hints with a structured, four-tier risk model. Module 11 makes the system’s reasoning auditable. Module 15 ensures the system improves continuously from physician feedback. All of these additions serve a single clinical goal: to give the physician an accurate, complete, and trustworthy pre-screening report before the consultation begins, so that consultation time is used for examination and clinical judgement rather than for basic information-gathering.

The empirical benchmarks of Phases 2 and 3 confirm that Module 1 must use a specialised, fine-tuned Bangla model (Qwen3-ASR-1.7B with LoRA, scheduled for CSE499B) rather than a general-purpose large model. The remainder of the fifteen modules will be designed, integrated, and validated in CSE499B.

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